Assignment 2

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data sent by Frank. We have a new transmission:

Mission Log 002 — from Frank

> Now Mission Control has actual *claims* they want checked. Like, official numbers someone wrote down before this mission even launched, and now they want to know: are those numbers still true, or did they just… write them down and hope?

Variable	Type	Description
`crew_role`	Categorical	Pilot, Engineer, Medic, or Analyst
`ship_module`	Categorical	Where the reading was logged: Engineering, Med Bay, Science Deck, Cargo Bay
`mission_phase`	Categorical	Transit, Orbit, or Planetside
`crew_sleep_hours`	Continuous	Hours slept in the last rest cycle
`hull_temp_f`	Continuous	Hull temperature (°F)
`o2_reserve_pct`	Continuous	Oxygen reserve remaining (%)

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's data here.

Question 1: Is Cinderspore growth accelerating?

The original mission brief assumed an average Cinderspore growth rate of 3.0 spores/hr. The crew is worried the real rate is higher than that.

1a. Before running any tests, construct the QQ plot for Cinderspore growth rate.

1b. Based on the graph above, do you believe that we meet the assumption for the one-sample *t*-test? You must explain why or why not.

Your answer here.

1c. Which approach will you use for inference? The one-sample *t*-test or the sign test? Why is that your choice?

Your answer here.

1d. Based on your answer in 1c, find the appropriate point and interval estimates for Cinderspore growth rate.

1e. Based on your answer in 1c, construct the appropriate hypothesis test to determine if the average Cinderspore growth rate is higher than 3.0 spores/hr.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if …

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude…

Question 2: Is the crew getting enough Engineering coverage?

Our staffing policy requires that at least 35% of logged crew activity come from Engineers, otherwise, the ship is considered under-resourced for repairs. The crew believes that Engineering coverage has slipped below that threshold.

2a. Find the appropriate point and interval estimates for Engineering's logged crew activity.

2b. Construct the appropriate hypothesis test to determine if Engineering coverage has slipped below 30%.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Question 3: Is the hull holding steady?

Our structural spec requires hull temperature to stay consistent: no more than 3°F of standard deviation is considered safe for long-term hull integrity. The crew wants to know if the hull temperature's variability is within spec, or if it's drifted beyond it.

3a. Find the appropriate point and interval estimates for the hull temperature.

3b. Construct the appropriate hypothesis test to determine if the hull temperature's variability has risen above 5°F.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

3c. What's the practical consequence for the crew if you determine the hull is not fine when it actually is?

Your answer here.

3d. What's the practical consequence for the crew if you determine the hull is fine when it actually is not?

Your answer here.